Formation-Preserving Distributed GNSS Spoofing Defeats Relative-Distance Detection in UAV Swarms

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[TODO: affiliazione, email]

***Abstract*—Multi-receiver spoofing detection is widely regarded as one of the most robust countermeasures against civilian GNSS spoofing: receivers deployed in a known formation compare the mutual distances implied by their GNSS fixes with the known (or independently measured) formation geometry, and a single-transmitter spoofer, which collapses all victims onto the same position, is detected with near-certain probability. In this paper we evaluate this detector family against a *distributed* spoofer: multiple ground transmitters deployed around the target area, each impersonating the satellite closest to it in azimuth. We develop a physical simulation model of a 25-UAV swarm overflown by 24 ground spoofers, including an elevation-dependent ground-to-UAV channel, and we implement the collapse test of Jansen et al. (ACSAC 2016) in its three variants together with a two-sided formation-consistency test representative of GNSS-vs-ranging checks. Our results quantify a *coordination gap*: the uncoordinated distributed attack distorts the swarm formation by 42–56 % of the baseline and is detected with probability up to 0.98 at a 10^{-2} false-alarm rate, whereas its coordinated counterpart—which imposes a common rigid translation on every receiver—preserves all pairwise distances exactly and reduces every detector in the family to chance level ($P_{\mathrm{D}} \approx P_{\mathrm{FA}}$ at every formation spacing from 5 to 100 m, ROC on the diagonal). The blindness is structural: any statistic computed from relative quantities only is invariant under a common translation. We position this result with respect to recent formal and experimental statements of the same limit, delimit its scope (direction-of-arrival and absolute-reference detectors are *not* blind to our ground configuration), and release the full MATLAB simulator.**

***Index Terms*—GNSS spoofing, UAV swarms, spoofing detection, multi-receiver, distributed attack, relative distances.**

I. Introduction

Civilian GNSS signals are unauthenticated and can be forged by commodity software-defined radios [1]. Among the proposed countermeasures, *multi-receiver* (or multi-agent) detection stands out for its simplicity: since a single-antenna spoofer forces every victim in its coverage area to compute the same position fix [2], a group of receivers that knows its own formation can detect the attack by checking that the mutual distances implied by the GNSS fixes remain consistent with the formation geometry [3], [4]. The same principle underpins recent proposals for UAV swarms, which compare GNSS-derived inter-drone distances against independent ultra-wideband (UWB) ranging [5], [6].

This paper asks whether the guarantee survives a *distributed* attacker, i.e. several coordinated ground transmitters surrounding the target area. Our contributions are: (i) a physical simulation model of a distributed ground spoofer attacking a moving UAV swarm, including per-satellite spoofer assignment and an elevation-dependent ground-to-UAV channel [7]; (ii) a faithful implementation of the collapse detector of Jansen et al. [3] and of a two-sided formation consistency test [8], [5]; (iii) a quantitative evaluation showing that the *uncoordinated* distributed attack is detectable, while its *coordinated* (rigid-translation) counterpart drives every detector of the family to chance level; and (iv) an honest delimitation of the result with respect to direction-of-arrival and absolute-reference defenses.

[TODO: aggiungere motivazione applicativa, struttura del paper]

II. Related Work

Multi-receiver position-domain detection: Tippenhauer et al. [2] formalized the GPS group-spoofing problem and proposed checking that the victims' computed positions preserve their physical formation; Jansen et al. [3] turned the check into a statistical test on pairwise distances with Rician/Rayleigh error models and spatially correlated noise, and Swaszek and Hartnett [4] derived Neyman–Pearson optimal variants for COTS receivers. Crowd-GPS-Sec [9] applies the same collapse principle to crowdsourced air-traffic receivers.

Measurement-domain and swarm variants: Stenberg et al. [8] test double differences of pseudorange and carrier phase across receivers (χ^2/GLRT), Kalantari and Larsson [10] derive a GLRT for receivers on a rigid body with unknown pose explicitly against multiple spoofers, and Borio and Gioia [11] detect the single common direction of arrival with a two-receiver sum-of-squares statistic. For UAV swarms, Mykytyn et al. [5] compare GNSS distances with UWB two-way ranging, and Bi et al. [6] cast detection as a semidefinite localization-feasibility problem. GNSS/INS fusion defenses such as SemperFi [12] and tightly coupled INS innovation monitors [13], [14] compare GNSS tracks with inertial dead reckoning.

Coordinated spoofing attacks: Result 7 of [2] proves that a formation-preserving group spoofing placement always exists (one transmitter per impersonated satellite); Jansen et al. [3] attempted a distance-preserving four-USRP attack and failed beyond one receiver. Ceccato et al. [15] designed a multi-antenna ground spoofer with Wiener precoding that displaces a swarm without disrupting its formation; Chen et al. [16] analyze formation-preserving spoofing of small

swarms; GNSS-WASP [17] demonstrated in hardware a transmitter constellation that spoofs receivers over a wide area while preserving their relative distances, explicitly bypassing multi-receiver [3] and sensor-fusion [12] countermeasures; and Park and Yoo [18] state the translation-invariance of relative-only detectors as a formal proposition, rooted in the gauge freedom of network localization [19]. Surveys of the field include [20], [21], [22].

[TODO: posizionamento rispetto a GNSS-WASP e Park–Yoo: il nostro contributo e' il modello fisico a terra e la quantificazione del coordination gap, non l'enunciato del limite]

III. System Model

A. Swarm, constellation, and measurement model

We consider a swarm of m UAVs flying in a rigid square grid of $\sqrt{m} \times \sqrt{m}$ elements with spacing g (the *formation spacing*), following a way-point trajectory at altitudes between 100 and 700 m; positions are sampled every $T_s = 5$ s. The GNSS constellation is modeled as $N_{\text{sat}} = 24$ satellites on 6 circular orbital planes (inclination $55°$, altitude 20 200 km) over a rotating spherical Earth. For UAV u at position $\mathbf{p}_u(t)$ and satellite s at position $\mathbf{s}_s(t)$ (visible when its elevation is positive), the nominal propagation delay is $\tau_{u,s}(t) = \|\mathbf{s}_s(t) - \mathbf{p}_u(t)\|/c$. Each receiver has a clock bias $\delta t_u \sim \mathcal{U}(-1, 1)$ s, constant over the run, and each pseudorange carries additive white Gaussian noise of standard deviation $\sigma_{\text{pr}} = 3$ m.

Each UAV computes its position fix $\hat{\mathbf{p}}_u$ independently by iterated linearized least squares (Gauss–Newton) over the four unknowns $(x, y, z, \delta t)$: at iteration k, with predicted ranges $R_s = \|\mathbf{s}_s - \hat{\mathbf{x}}^{(k)}\|$ and pseudoranges $\rho_s = c\,(\tilde{\tau}_{u,s} - \widehat{\delta t}^{\,(k)})$,

$$\mathbf{G}\,\Delta = \boldsymbol{\rho} - \mathbf{R}, \qquad \mathbf{G}_{s,:} = \left[\frac{(\hat{\mathbf{x}} - \mathbf{s}_s)^\top}{R_s},\ c\right], \tag{1}$$

solved in the least-squares sense for five iterations, where $\tilde{\tau}_{u,s}$ is the *measured* delay (authentic or spoofed).

In addition to GNSS, the UAVs measure their true mutual distances with an independent on-board ranging sensor (e.g. UWB two-way ranging [5]):

$$d_{ij}^{\text{meas}}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| + \varepsilon_{ij}(t), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{\text{rng}}^2), \tag{2}$$

with $\sigma_{\text{rng}} = 0.1$ m. This is the most favorable setting for the defender: the formation geometry is known essentially without error.

B. Ground-to-UAV channel

Spoofing signals travel over a ground-to-air link. We adopt the suburban cellular-to-UAV path-loss model of Al-Hourani and Gomez [7]: for a ground transmitter at distance d and depression/elevation angle θ (in degrees) as seen from the UAV,

$$\text{PL}(d, \theta) = 10\,\alpha \log_{10}(d) + A\,(\theta - \theta_0)\,e^{-\frac{\theta - \theta_0}{B}} + \eta_0 \quad [\text{dB}], \tag{3}$$

with $\alpha = 3.04$, $A = -23.29$, $B = 4.14$, $\theta_0 = -3.61°$, $\eta_0 = 20.70$ dB (shadowing disabled for repeatability). Authentic signals follow free-space loss from satellite transmit power $P_t^{\text{sat}} = 25$ W with antenna gains $G_t^{\text{sat}} = 13.5$ dB and $G_r = 5$ dB.

IV. Distributed Spoofing Attack

The attacker's goal is to displace the whole swarm's perceived position onto a target trajectory (a fixed decoy point in our runs, roughly 5.7 km from the flight path), while the swarm believes it is following its mission plan.

A. Transmitter geometry and satellite assignment

$N_{\text{sp}} = 24$ ground transmitters are placed on a circle of radius $R_{\text{sp}} = 2.5$ km centered on the target area (Fig. 1), each radiating $P_t^{\text{sp}} = 1$ mW with $G_t^{\text{sp}} = 1$ dB. Following the geometric feasibility argument of [2, Result 7]—a formation-preserving group spoofing placement uses one transmitter per impersonated satellite—each visible satellite s is assigned to the spoofer whose azimuth, as seen from the area center, is closest to the satellite's azimuth:

$$\text{sp}(s, t) = \arg\min_k \left|\angle_{2\pi}\big(\phi_s(t) - \phi_k\big)\right|, \tag{4}$$

where ϕ_s is the satellite azimuth and ϕ_k the azimuth of spoofer k.

B. Uncoordinated attack (*power*)

In the baseline attack the spoofers transmit a common signal set, with no per-victim adaptation. For each UAV u and satellite s, the receiver locks onto the spoofer whenever the received spoofing power exceeds the authentic one, $P_r^{\text{sp}} > P_r^{\text{sat}}$, with P_r^{sp} given by (3). The measured delay is then

$$\tilde{\tau}_{u,s} = \tau_s^{\text{dec}} - \delta t_u - \frac{R_{\text{sp}} - d_{u,\text{sp}(s)}}{c}, \tag{5}$$

where τ_s^{dec} is the delay that satellite s would exhibit at the decoy position and $d_{u,\text{sp}(s)}$ is the true spoofer-to-UAV distance. The last term is the physical propagation from the assigned spoofer: it varies across the swarm along the *horizontal* direction towards the spoofer, whereas an authentic delay varies along the inclined line of sight to the satellite. This mismatch makes the induced position offset *position-dependent*, so the swarm is displaced but also *deformed*.

C. Coordinated attack (*rigid*)

The coordinated attacker knows the swarm geometry (or equivalently, adapts per victim) and pre-compensates the per-UAV terms so that every receiver measures delays consistent with its *true* position shifted by a common vector:

$$\tilde{\tau}_{u,s} = \frac{\|\mathbf{s}_s - (\mathbf{p}_u + \boldsymbol{\Delta})\|}{c} - \delta t_u, \qquad \boldsymbol{\Delta}(t) = \mathbf{p}^{\text{dec}}(t) - \mathbf{p}^{\text{ctr}}(t), \tag{6}$$

where $\mathbf{p}^{\text{ctr}}$ and $\mathbf{p}^{\text{dec}}$ are the true and decoy trajectories of the swarm center. Solving (1) then yields $\hat{\mathbf{p}}_u = \mathbf{p}_u + \boldsymbol{\Delta} + \mathbf{n}_u$: the whole swarm is rigidly translated. The transmitters are *not* moved; (6) models the delays that the coordinated ground array must synthesize, i.e. the ideal end state of the attack of [2], [17]. The gap between (5) and (6) is precisely the coordination effort required of the attacker, and Section VI quantifies its observable consequence.

D. Why rigid translations are undetectable

Proposition 1 (Translation invariance): Let $\hat{\mathbf{p}}_u = \mathbf{p}_u + \boldsymbol{\Delta} + \mathbf{n}_u$, $u = 1, \dots, m$, with $\boldsymbol{\Delta}$ common to all receivers, and let the detector statistic be any measurable function

$$T = f\big(\{\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\}_{i<j},\, \{d_{ij}^{\text{meas}}\}_{i<j}\big). \tag{7}$$

Then T has the same distribution under H_0 ($\boldsymbol{\Delta} = \mathbf{0}$) and under the rigid attack ($\boldsymbol{\Delta} \neq \mathbf{0}$); consequently $P_{\text{D}} = P_{\text{FA}}$ for every threshold.

Proof: $\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j = (\mathbf{p}_i - \mathbf{p}_j) + (\mathbf{n}_i - \mathbf{n}_j)$ does not depend on $\boldsymbol{\Delta}$, and d_{ij}^{meas} is measured independently of GNSS. Hence the joint distribution of the arguments of f, and therefore of T, is identical under the two hypotheses. ∎

If the statistic uses only distances (not vectors), invariance extends to the whole Euclidean group (common rotations and reflections). The proposition restates, in the spoofing context, the gauge freedom of anchor-free network localization [19]; it appears implicitly in [2, Result 7] and explicitly in [18], and is realized in hardware by [17]. Our contribution is not the statement but its quantitative confrontation with a *physical ground-based* attacker model.

V. Relative-Distance Detection Methods

All detectors operate on the normalized distance ratios

$$r_{ij}(t) = \frac{\hat{d}_{ij}(t)}{d_{ij}^{\text{meas}}(t)}, \qquad \hat{d}_{ij} = \|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\|, \tag{8}$$

computed over all $\binom{m}{2} = 300$ pairs at every epoch; under authentic conditions $r_{ij} \approx 1$.

A. Collapse tests (Jansen et al.)

The ACSAC 2016 detector [3] declares spoofing when the pairwise distances *shrink*—the signature of a single-transmitter spoofer that collapses all fixes onto one point:

$$f\big(\{r_{ij}\}\big) \underset{H_1}{\overset{H_0}{\gtrless}} \lambda, \qquad \text{alarm if } f < \lambda, \tag{9}$$

with the three variants of its Appendix A implemented here:

$$f_{\min} = \min_{i<j} r_{ij}, \quad f_{\max} = \max_{i<j} r_{ij}, \quad f_{\text{maj}} = \operatorname*{med}_{i<j} r_{ij}, \tag{10}$$

where the median form is equivalent to the paper's (normalized) majority voting: at least half of the pairs below λ iff the median is. Jansen et al. set λ from a Rician (H_0) / Rayleigh (H_1) model of the pairwise distance with spatially correlated position noise, $\sigma_d = \sqrt{2}\,\sigma\sqrt{1-\rho}$; we calibrate λ empirically to the same effect (Sec. V-C).

B. Formation consistency test

As a representative of the two-sided GNSS-vs-ranging consistency family — $|d_{\text{GPS}} - d_{\text{UWB}}| > d_{\text{thr}}$ for UAV swarms [5], double-difference GLRTs across receivers [8], GLRTs fusing GNSS fixes with inter-vehicle ranges [23] — we use the worst-pair statistic

$$T_{\text{cons}} = \max_{i<j} \big|\, r_{ij} - 1 \,\big|, \qquad \text{alarm if } T_{\text{cons}} > \lambda. \tag{11}$$

TABLE I
Simulation parameters.

Parameter	Value
Swarm size m	25 (5×5 grid)
Formation spacing g	5–100 m
Mission duration / epoch	1200 s / $T_s = 5$ s
Satellites	24 (6 planes, 55°, 20 200 km)
Pseudorange noise σ_{pr}	3 m
Ranging noise σ_{rng}	0.1 m
Ground spoofers N_{sp}	24 on circle, $R_{\text{sp}} = 2.5$ km
Spoofer EIRP	1 mW + 1 dB
Ground-to-UAV channel	Eq. (3), [7]
Target false-alarm rate P_{FA}	10^{-2}
Monte Carlo runs	2 per (spacing, scenario)

Unlike (9), this test fires on *any* deformation of the formation, in either direction, and is therefore the natural candidate against a distributed attacker that scatters rather than collapses the fixes.

C. Threshold calibration and metrics

Following the Neyman–Pearson practice of [3], thresholds are calibrated on the H_0 (no-spoofing) scenario, simulated with the same noise levels: λ is the empirical P_{FA}-quantile (lower quantile for the collapse tests, upper for the consistency test) of the statistic over all H_0 epochs, with $P_{\text{FA}} = 10^{-2}$. Performance is then reported as (i) the detection probability P_{D}, the fraction of attacked epochs that raise an alarm; (ii) full ROC curves obtained by sweeping λ; and (iii) the median relative distance distortion $E = \operatorname{med}_{i<j} |\hat{d}_{ij} - d_{ij}|/d_{ij}$, which measures the geometric evidence available to any relative-distance detector.

VI. Simulation Results

A. Setup

Table I summarizes the simulation parameters. For each formation spacing $g \in \{5, 10, 20, 50, 100\}$ m and each scenario (H0, rigid, power) we run $N_{\text{MC}} = 2$ Monte Carlo repetitions of the full 20-minute mission (240 epochs each); the random start time, clock biases, and noise realizations are shared across scenarios within a repetition, so differences are attributable to the attack alone. The simulator and detection code are available in the companion repository.[1]

B. Attack effectiveness

Both attacks succeed at displacing the swarm: under rigid the center UAV's estimated position is dragged by the full decoy offset (4.0 km on average over the mission) at every epoch, and under power a comparable displacement (4.6 km on average) is achieved (Fig. 2, top). The difference is in what the attacks do to the swarm's *shape* (Fig. 2, bottom): the uncoordinated attack deforms the formation persistently, while the coordinated attack leaves the relative geometry statistically indistinguishable from the no-attack case.

[1] https://github.com/silviasch01/GPS_spoofing_distribuito

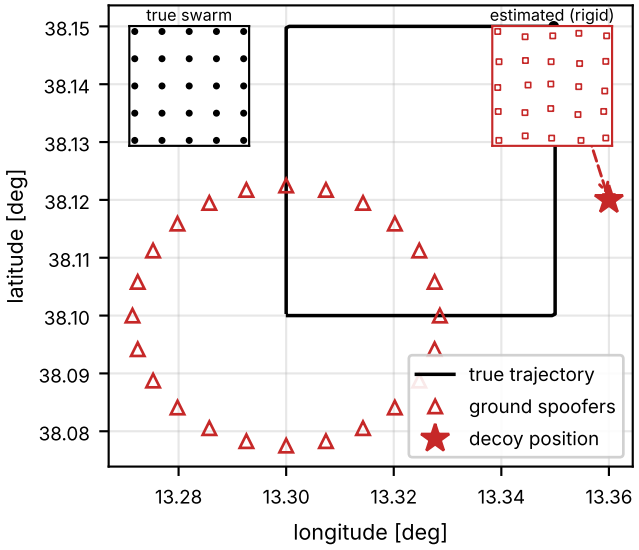


Fig. 1. Attack scenario (one epoch, $g = 20\,\mathrm{m}$). The swarm (true positions, dots) flies its mission; 24 ground spoofers (triangles) on a 2.5 km circle impersonate one satellite each. Under the coordinated attack the estimated fixes (open squares) reproduce the formation rigidly translated onto the decoy trajectory (dashed).

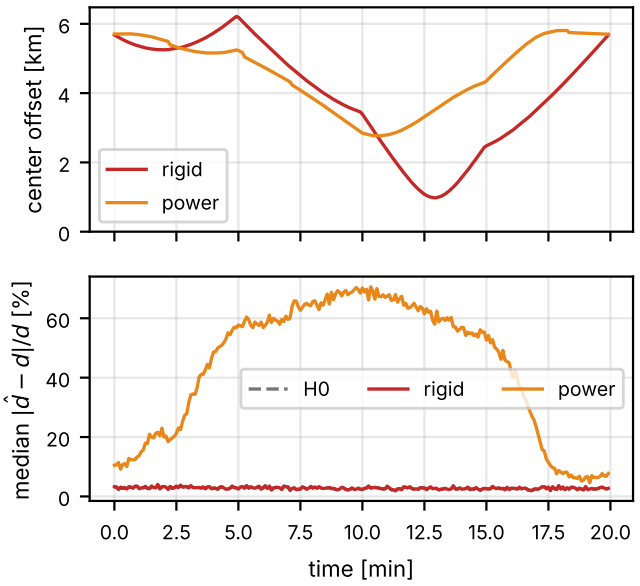


Fig. 2. Per-epoch behaviour at $g = 20\,\mathrm{m}$ (single run). Top: position offset of the center UAV—both attacks displace the swarm by kilometres. Bottom: median relative-distance error—rigid (red) coincides with H0 (grey), power (orange) deforms the formation by $\sim$50%.

C. Formation distortion

Figure 3 reports the median relative distance distortion E. Under H0 the distortion is pure GNSS noise and decays as $1/g$ (from 18 % at $g = 5\,\mathrm{m}$ to 0.8 % at 100 m, with $\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$). The rigid attack is *identical* to H0 to the fourth significant digit at every spacing—the common translation cancels exactly

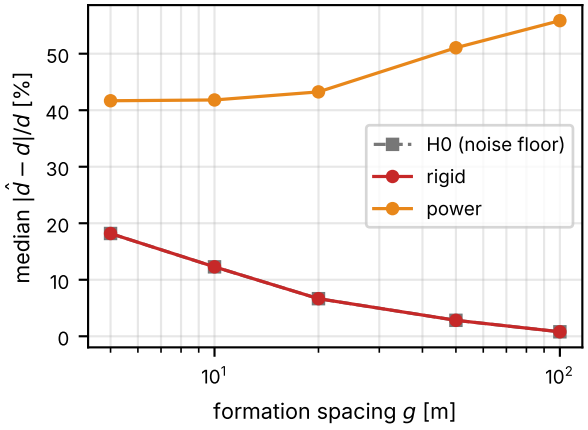


Fig. 3. Median relative-distance distortion vs. formation spacing. rigid coincides with the no-attack noise floor; power deforms the formation by $\sim$50% at every spacing.

TABLE II
DETECTION PROBABILITY AT $P_{\mathrm{FA}} = 10^{-2}$ ($\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$, $\sigma_{\mathrm{rng}} = 0.1\,\mathrm{m}$; 480 EPOCHS PER CELL). H0 COLUMNS VERIFY THE CALIBRATION.

	Jansen majority (10)			Consistency (11)		
g [m]	H0	rigid	power	H0	rigid	power
5	0.008	**0.010**	0.733	0.008	**0.008**	0.010
10	0.008	**0.010**	0.754	0.008	**0.010**	0.025
20	0.008	**0.010**	0.667	0.008	**0.010**	0.196
50	0.008	**0.010**	0.792	0.008	**0.008**	0.733
100	0.008	**0.008**	0.790	0.008	**0.008**	0.975

in every pairwise difference, as predicted by Proposition 1. The power attack instead distorts the formation by 42–56 % regardless of spacing: the per-UAV propagation term in (5) scales with the baseline, so the deformation never averages out and, for $g \gtrsim 20\,\mathrm{m}$, towers over the noise floor.

D. Detection performance

Table II and Fig. 4 report P_{D} at $P_{\mathrm{FA}} = 10^{-2}$; Fig. 5 shows the ROC curves at $g = 20\,\mathrm{m}$. Three observations follow.

First, the **uncoordinated attack is detectable**. The consistency test (11) reaches $P_{\mathrm{D}} = 0.73$ at $g = 50\,\mathrm{m}$ and 0.98 at 100 m, where the 50 % deformation dwarfs the noise; below 20 m the noise floor masks it. The Jansen variants detect power only partially ($P_{\mathrm{D}} \approx 0.65$–$0.80$): they are one-sided collapse tests, and the distributed attack *scatters* the fixes instead of collapsing them—indeed $f_{\max}$, which requires every pair to shrink, barely fires at all ($P_{\mathrm{D}} \leq 0.66$).

Second, the **coordinated attack is invisible**: for all four detectors and all spacings, $P_{\mathrm{D}} \in \{0.008, 0.010\} \approx P_{\mathrm{FA}}$, and the ROC curves for rigid lie on the chance diagonal (Fig. 5). This is not a tuning artifact: by Proposition 1 no threshold, window length, or variant of a relative-only statistic can do better.

Third, the comparison quantifies the **coordination gap**: the 42–56 % residual deformation of (5) is what currently separates a physical ground array from the null space. An

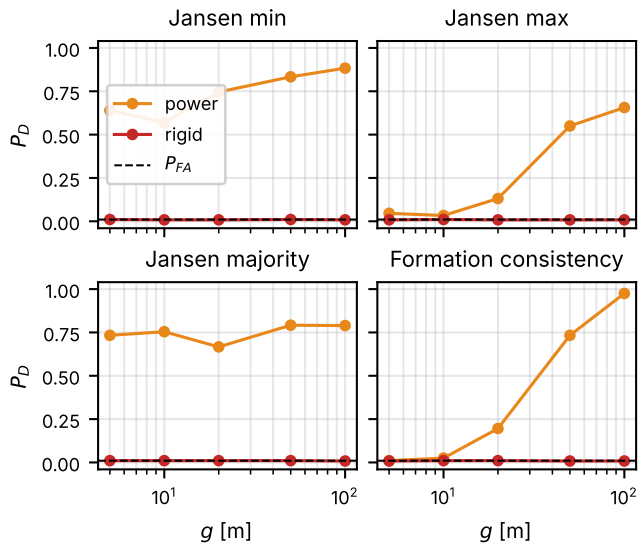


Fig. 4. Detection probability vs. formation spacing at $P_{\mathrm{FA}} = 10^{-2}$ for the four detectors. The rigid curves lie on the P_{FA} line at every spacing.

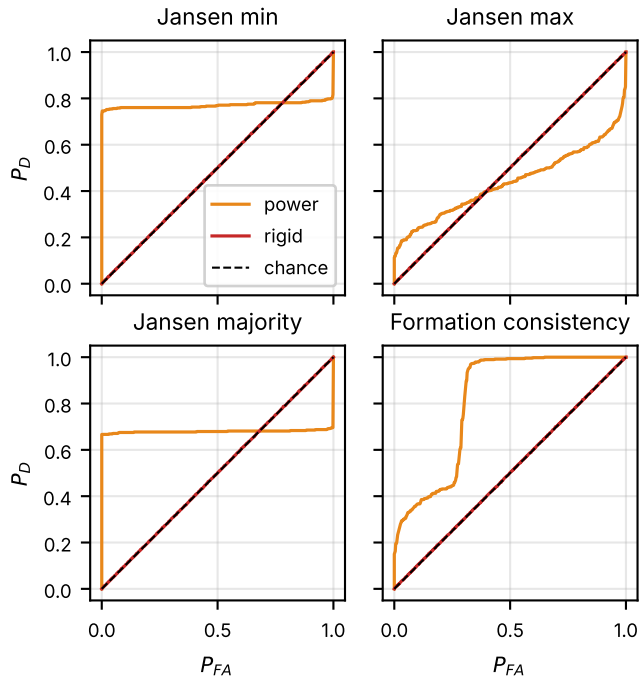


Fig. 5. ROC curves at $g = 20\,\mathrm{m}$. Under the coordinated attack (rigid) every detector operates on the chance diagonal.

attacker able to apply the per-victim pre-compensation of (6)—which requires knowledge of each UAV's position to within the detector's noise floor, i.e. metres—closes the gap entirely.

VII. Discussion and Limitations

Scope of the blindness claim: Proposition 1 covers detectors that are functions of relative position/distance quantities—the collapse and consistency families evaluated here, the double-difference and rigid-body GLRTs of [8], [10], the swarm GNSS-vs-UWB checks of [5], [6]—and, in the time domain, GNSS/INS detectors built on displacement consistency: the maneuver statistic of SemperFi [12] is computed on displacements from the maneuver start, so a constant $\boldsymbol{\Delta}$ cancels identically, and in tightly coupled INS monitors a constant offset absorbed at filter initialization produces zero innovation [13], [14]. It does *not* cover (i) *direction-of-arrival* detectors: a circle of ground transmitters produces arrivals at near-zero elevation, incongruent with the satellite constellation, so multi-antenna arrays with attitude information would detect our attacker unless its transmitters are placed along the satellite directions as in [17]; nor (ii) *absolute-reference* checks—ranging to fixed infrastructure, TDoA from ground sensor networks [9], terrain or signal-of-opportunity positioning—for which the common translation is directly observable. A complete defense should therefore anchor at least one absolute reference, consistent with the anchor requirements derived in [18].

Idealized rigid model: Eq. (6) prescribes the delays of a *perfectly* coordinated array; it is an upper bound for the attacker, adopted deliberately so that the detector-side conclusion is threshold-free. A physical ground array approximates it with a residual non-rigidity that grows with swarm aperture and shrinks with transmitter altitude and range—GNSS-WASP measured sub-metre residuals at 1 km in hardware [17], below every published formation threshold (e.g. $\approx$5 m [5], $\approx$3.6 m [3]). Mapping the residual of *our* ground-circle geometry as a function of radius, altitude, and swarm extent is the natural next step; the power results already bound it from above.

Other simplifications: The simulator uses a spherical Earth, a synthetic constellation, uncorrelated pseudorange noise (spatial correlation would *lower* the detector's noise floor [3], strengthening the power detection result but leaving Proposition 1 untouched), no multipath, and signal capture by simple power comparison without tracking-loop dynamics.

VIII. Conclusion

We evaluated the classical multi-receiver relative-distance spoofing detectors against a distributed ground spoofer attacking a UAV swarm. An uncoordinated 24-transmitter attack displaces the swarm but deforms its formation by $\sim$50% and is detected (up to $P_{\mathrm{D}} = 0.98$ at $P_{\mathrm{FA}} = 10^{-2}$); the same attack with per-victim coordination imposes a common rigid translation, preserves every pairwise distance, and reduces the entire detector family—Jansen's collapse variants and the formation-consistency test—to chance level at every formation spacing, as dictated by a simple invariance argument. The practical security of relative-distance detection thus rests not on its statistic but on the attacker's coordination cost, which recent hardware results are rapidly eroding; robust swarm navigation requires anchoring to at least one absolute position reference.

Acknowledgments

[TODO: ringraziamenti, funding]

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